

Phase 1: Python programming (10 hours)

Week 1: Python Basics & Core Programming Concepts

1 Python Fundamentals

- Setting up Python Environment (Jupyter Notebook, VS Code, Anaconda)
- Python Data Types: Integers, Floats, Strings, Booleans
- Variables and Memory Management
- Operators: Arithmetic, Comparison, Logical, Bitwise
- Type Casting & String Manipulation

2 Data Structures & Control Flow

- Lists, Tuples, Sets, Dictionaries (Usage in data processing)
- Loops: for, while, and Iterators
- Conditional Statements: if-elif-else
- List & Dictionary Comprehensions (Efficient Data Processing)
- Functions:
- Defining & Calling Functions
- *args and **kwargs
- Lambda Functions

3 Working with Files & Exception Handling

- Reading & Writing Files (CSV, JSON, TXT)
- Handling Exceptions (try-except, finally)
- Logging in Python

Hands-on Exercise

Week 2: NumPy, Pandas, and Data Handling

4 NumPy for Numerical Computations

- Creating NumPy Arrays (ndarray, dtype, shape)
- Indexing, Slicing, and Broadcasting
- Mathematical Operations (Dot Product, Matrix Multiplication)
- Handling Missing Data in Arrays
- Statistical Operations (mean, std, var, percentile)

5 Pandas for Data Manipulation

- Loading Datasets (read_csv, read_json, read_excel)
- Pandas Series vs. DataFrame (Data Structures)
- DataFrame Operations:
- Indexing, Filtering, Sorting
- GroupBy & Aggregations (sum, mean, count, apply)
- Handling Missing Data (dropna, fillna, interpolate)
- Merging, Concatenation & Joins in Pandas

Hands-on Exercise

Week 3: Data Visualization, APIs, and Automation

6 Data Visualization with Matplotlib & Seaborn

- Matplotlib Basics: Line, Bar, Scatter, and Pie Charts
- Seaborn for Statistical Visualization:
- Histograms, KDE plots
- Boxplots & Violin Plots (Detecting Outliers)
- Heatmaps (Correlation Matrices)

- Customization: Legends, Colors, Titles, Annotations

7 Working with APIs & Web Scraping

- Making API Calls using requests
- Handling JSON responses
- Introduction to Web Scraping (BeautifulSoup, Scrapy)
- Automating Data Collection

8 Regular Expressions (Regex) for Text Processing

- Pattern Matching & String Manipulation (re.search, re.findall)
- Cleaning Text Data (Removing Special Characters, Tokenization)
- Applying Regex in Pandas

Hands-on Exercise

Phase 2: Statistics & Probability (15 hours)

Week 4: Descriptive Statistics & Data Understanding

1 Understanding Data Types & Distributions

- Types of Data: Categorical vs. Numerical (Discrete & Continuous)
- Measures of Central Tendency: Mean, Median, Mode
- Measures of Dispersion: Variance, Standard Deviation, Range, IQR
- Skewness & Kurtosis: Identifying asymmetry and shape of distributions
- Data Visualization:
- Histograms & Boxplots (for detecting outliers)
- KDE (Kernel Density Estimation) plots
- Scatter plots & Pair plots (Exploring relationships)

2 Outliers & Data Cleaning

- Z-score & Standardization
- IQR (Interquartile Range) Method for Outlier Detection
- Winsorization & Transformation Techniques (Log, Square Root, etc.)

Hands-on Exercise

Weeks 5 and 6: Probability Theory & Inferential Statistics

3 Probability Foundations

- Basic Probability Rules (Addition, Multiplication, Conditional Probability)
- Bayes' Theorem & Applications (Spam detection, Medical diagnosis, etc.)
- Probability Distributions:
- Discrete Distributions: Bernoulli, Binomial, Poisson
- Continuous Distributions: Uniform, Normal, Exponential
- Central Limit Theorem (CLT): Importance in large-scale data analysis

4 Hypothesis Testing & Confidence Intervals

- Statistical Hypothesis Testing
- Null Hypothesis (H_0) & Alternative Hypothesis (H_a)
- p-value & Significance Level (α)
- Types of Hypothesis Tests:
- Z-test (for large samples, known variance)
- T-test (for small samples, unknown variance)
- Chi-square test (for categorical data independence)

- ANOVA (Comparing multiple groups)
- Confidence Intervals:
- How to interpret confidence intervals?
- Bootstrapping for interval estimation

Hands-on Exercise

Phase 3: Classical Machine Learning (25 hours)

Week 1: Introduction to Machine Learning & Supervised Learning

- 1 Machine Learning Basics
 - What is Machine Learning?
 - Types of ML:
 - Supervised (Regression, Classification)
 - Unsupervised (Clustering, Dimensionality Reduction)
 - Reinforcement Learning (Basic Introduction)
 - ML Workflow: Data Preprocessing → Model Training → Model Evaluation
- 2 Linear Regression (Predicting Continuous Variables)
 - Simple Linear Regression (One Predictor)
 - Multiple Linear Regression (Multiple Predictors)
 - Assumptions of Linear Regression (Linearity, No Multicollinearity, Homoscedasticity)
 - Feature Selection Techniques:
 - Recursive Feature Elimination (RFE)
 - LASSO & Ridge Regularization
 - Evaluating Regression Models:
 - RMSE, MAE, R^2 Score

Hands-on Exercise

Week 2: Classification Models & Model Evaluation

- 3 Logistic Regression (Binary Classification)
 - Sigmoid Function & Decision Boundary
 - Cost Function for Classification
 - ROC Curve, AUC, Precision-Recall Tradeoff
 - Handling Class Imbalance (Oversampling & Undersampling, SMOTE)
- 4 Decision Trees & Random Forests
 - How Decision Trees Work (Gini Index, Entropy, Information Gain)
 - Overfitting & Pruning in Decision Trees
 - Random Forests for Ensemble Learning
 - Feature Importance in Tree-based Models
- 5 Support Vector Machines (SVMs)
 - How SVM Works: Finding the Optimal Hyperplane
 - Kernel Trick (Linear, Polynomial, RBF Kernels)
 - Tuning C and Gamma for Model Performance

Hands-on Exercise

Week 3: Unsupervised Learning & Dimensionality Reduction

- 6 Clustering Techniques (Grouping Similar Data)
 - K-Means Clustering:
 - Choosing the Optimal K (Elbow Method, Silhouette Score)
 - Hierarchical Clustering
 - DBSCAN (Density-Based Clustering)
- 7 Principal Component Analysis (PCA)
 - Reducing Dimensions to Improve Model Efficiency
 - Eigenvectors & Eigenvalues Intuition
 - Explained Variance & Selecting the Right Number of Components
- 8 Feature Engineering & Data Preprocessing
 - Handling Missing Values (Mean/Median Imputation, KNN Imputation)
 - Encoding Categorical Variables (One-Hot Encoding, Label Encoding, Target Encoding)
 - Feature Scaling (Standardization, Min-Max Scaling, Robust Scaling)
 - Feature Extraction Techniques

Hands-on Exercise

Week 4: Advanced ML Concepts & Hyperparameter Tuning

- 9 Gradient Boosting Algorithms (XGBoost, LightGBM, CatBoost)
 - Why Boosting Works?
 - XGBoost & LightGBM Optimization Techniques
 - Regularization in Boosting Models
- 10 Model Selection & Hyperparameter Tuning
 - Grid Search vs. Random Search vs. Bayesian Optimization
 - Automated ML Tuning using Optuna
 - Cross-Validation Techniques (K-Fold, Stratified K-Fold, Leave-One-Out)

Hands-on Exercise

Week 5: Time Series Forecasting & Real-World Applications

- 1 1 Time Series Forecasting
 - Introduction to Time Series Data
 - Stationarity & Differencing Techniques
 - ARIMA, SARIMA, and Prophet Models
- 1 2 Model Deployment & Interpretability
 - Deploying ML Models using Flask/FastAPI
 - Explainable AI (SHAP, LIME for Feature Interpretability)

Hands-on Exercise

Phase 4: Deep Learning (20 hours)

Week 1: Neural Networks & Deep Learning Foundations

- 1 Introduction to Deep Learning & Neural Networks
 - Why Deep Learning?
 - Comparison: ML vs. Deep Learning vs. AI
 - Biological Neurons vs. Artificial Neurons
 - Activation Functions: ReLU, Sigmoid, Tanh, Softmax
 - Forward & Backpropagation in Neural Networks
 - Loss Functions (MSE, Cross-Entropy)
- 2 Building Feedforward Neural Networks (FNNs)
 - Understanding Weights & Biases
 - Gradient Descent & Optimizers (SGD, Adam, RMSprop)
 - Overfitting & Regularization: Dropout, L1/L2 Regularization
 - Batch Normalization for Stable Training

Hands-on Exercise

Week 2: Convolutional Neural Networks (CNNs) for Computer Vision

- 3 CNN Architecture & Applications
 - Why CNNs for Image Data?
 - Convolution Operation & Filters
 - Stride, Padding, and Pooling Layers
 - Flattening & Fully Connected Layers
- 4 Building & Training CNN Models
 - Popular Architectures: LeNet, AlexNet, VGG, ResNet
 - Transfer Learning with Pretrained Models
 - Data Augmentation & Handling Imbalanced Datasets

Hands-on Exercise

Week 3: Recurrent Neural Networks (RNNs)

- 5 Sequence Modeling & Recurrent Networks
 - Why RNNs for Sequential Data?
 - Vanishing Gradient Problem & LSTMs/GRUs

Hands-on Exercise

Week 4: NLP

- 1 Introduction to NLP
 - What is NLP?
 - Real-world applications (Search engines, chatbots, sentiment analysis)
 - Challenges in NLP
 - Ambiguity, Context Understanding, Sarcasm
 - Hands-on:
 - Implementing basic text preprocessing
- 2 Text Preprocessing Techniques

- Tokenization
 - Word & Sentence Tokenization
- Text Cleaning
 - Stopword Removal, Lemmatization vs. Stemming
- Hands-on:
 - Preprocessing text using NLTK & spaCy

3 Text Representation Techniques

- Traditional Approaches
 - Bag of Words (BoW) & TF-IDF
- Word Embeddings
 - Word2Vec, GloVe, FastText
- Contextual Embeddings
 - BERT, GPT
- Hands-on:
 - Implementing embeddings using Scikit-learn & Gensim

4 Named Entity Recognition (NER) & POS Tagging

- POS Tagging & Dependency Parsing
 - Understanding syntactic structures
- Named Entity Recognition (NER)
 - Using spaCy for entity extraction
- Hands-on:
 - Extracting key entities from resumes

5 Sequence Modeling & Recurrent Networks

- Why RNNs for Sequential Data?
 - Time-series & language modeling use cases
- Challenges with RNNs
 - Vanishing Gradient Problem
- LSTMs & GRUs
 - How they solve RNN limitations
- Hands-on:
 - Training an LSTM for sentiment analysis

6 Sentiment Analysis & Text Classification

- Traditional Machine Learning Approaches
 - Naive Bayes, SVM for text classification
- Deep Learning Approaches
 - Transformer-based Sentiment Analysis (BERT, DistilBERT)
- Hands-on:
 - Fine-tuning BERT for text classification

7 Text Similarity & Semantic Search

- Text Similarity Metrics
 - Cosine Similarity, Jaccard Similarity
- Semantic Search
 - Using BERT & SBERT
- Hands-on:
 - Implementing SBERT for FAQ retrieval

8 Machine Translation & Text Generation

- Sequence-to-Sequence Models
 - LSTMs, Transformers for translation
- Neural Machine Translation (NMT)

- Using MarianMT for multilingual translation
- Hands-on:
 - English-to-French translation with Hugging Face

9 Topic Modeling & Text Summarization

- Topic Modeling Techniques
 - LDA (Latent Dirichlet Allocation)
- Summarization Approaches
 - Extractive (TextRank), Abstractive (BART)
- Hands-on:
 - Summarizing long articles with BART

Phase 5: Generative AI and Prompting techniques (20 hours)

PART 1: FUNDAMENTALS OF GENERATIVE AI

1 Introduction to Generative AI

- What is Generative AI?
 - Difference between Discriminative & Generative Models
- Real-world Applications of Generative AI
 - Image generation, text generation, code generation, music synthesis
- Hands-on:
 - Exploring OpenAI's GPT and DALL·E models

2 Foundation Models in Generative AI

- Overview of Large Language Models (LLMs)
 - GPT, BERT, T5, LLaMA, Mistral
- Pre-training & Fine-tuning of LLMs
 - Supervised Fine-tuning vs. Reinforcement Learning from Human Feedback (RLHF)
- Hands-on:
 - Fine-tuning GPT-3.5/4 with domain-specific datasets

3 Transformer Architecture & Mechanisms

- Understanding Transformer-based Models
 - Self-attention, positional encoding, multi-head attention
- Differences between Auto-regressive & Auto-encoding Models
 - GPT vs. BERT vs. T5
- Hands-on:
 - Implementing a Transformer from scratch in PyTorch

◆ PART 2: PROMPT ENGINEERING & TECHNIQUES

4 Introduction to Prompting Techniques

- What is Prompt Engineering?
- Importance of Effective Prompts in AI Models
- Hands-on:
 - Experimenting with basic prompts using OpenAI's API

5 Types of Prompting Techniques

- Zero-shot Prompting
 - Generating responses without providing examples
- Few-shot Prompting

- Providing contextual examples to guide AI behavior
- Chain-of-Thought (CoT) Prompting
 - Step-by-step reasoning for complex problem-solving
- Hands-on:
 - Testing different prompting techniques for text generation

6 Advanced Prompt Engineering Strategies

- Self-consistency Prompting
 - Improving reliability in AI-generated responses
- Role-based & Persona-based Prompting
 - Using AI to mimic specific styles & behaviors
- Hands-on:
 - Designing prompts for creative writing & chatbots

7 Prompt Optimization & Debugging

- Prompt Refinement Strategies
- Handling Model Hallucinations & Biases
- Hands-on:
 - Debugging AI-generated outputs with iterative prompt tuning

◆ PART 3: GENERATIVE AI APPLICATIONS

8 Text Generation & AI Writing Assistants

- Natural Language Generation (NLG)
- AI for Content Creation & Summarization
- Hands-on:
 - Generating blog articles & summaries using GPT

9 Image Generation with Diffusion Models

- Understanding Diffusion Models (Stable Diffusion, DALL·E)
- Customizing AI-generated images with control prompts
- Hands-on:
 - Generating and editing AI images using prompt-based techniques

10 Multimodal AI & Interactive Applications

- Combining Text, Images, and Audio in AI Models
- Understanding CLIP (Contrastive Language-Image Pretraining)
- Hands-on:
 - Implementing multimodal AI using CLIP

11 AI in Code Generation & Productivity Tools

- AI-powered Code Assistants (GitHub Copilot, Code Llama)
- AI-driven Workflow Automation
- Hands-on:
 - Using AI for code generation and debugging

◆ PART 4: ETHICS, DEPLOYMENT & PROJECTS

12 Ethical Considerations in Generative AI

- Bias, Misinformation & Hallucinations in AI
- Copyright & Legal Issues in AI-generated Content
- Hands-on:

- Evaluating AI model bias with real-world examples

1 3 Deploying Generative AI Models

- Model Deployment Strategies
 - APIs, Cloud Deployment, Containers
- Optimizing AI Model Performance & Cost
- Hands-on:
 - Deploying a GPT-based chatbot using FastAPI

Phase 6: Ancillary Skills for AI & ML Practitioners (10 hours)

Week 1: Big Data Processing & Distributed Computing

1 Introduction to Big Data for AI

- Why Big Data is Important for AI/ML?
- Challenges in Handling Large-Scale AI Workloads
- Big Data Ecosystem: Spark, and Dask

2 Scalable Data Processing for AI Pipelines

- Apache Spark for Data Processing
- Pandas vs. Dask vs. Spark for ML Workflows
- ETL Pipelines for AI: Batch vs. Streaming Data Processing

Hands-on Exercise

Week 2: Cloud Computing & AI Workflows

3 Cloud AI Services & Model Deployment

- Overview of Cloud AI Platforms: AWS, GCP, Azure
- AutoML & Managed AI Services (AWS SageMaker, GCP Vertex AI)
- Scaling AI Models with Cloud GPUs & TPUs

4 Serverless & Containerized AI Deployments

- Dockerizing AI Applications for Portability
- Kubernetes for AI Model Scaling
- Deploying AI Models as Serverless Functions (AWS Lambda, GCP Cloud Functions)

Hands-on Exercise

Week 3: ML Engineering & Software Development Best Practices

5 ML Engineering & Best Coding Practices

- Structuring ML Code for Maintainability
- Modularization, OOP & Functional Programming in AI
- Unit Testing for AI Codebases (pytest, unittest)

6 CI/CD Pipelines for AI Models

- Introduction to CI/CD for ML (Continuous Integration & Deployment)
- Using GitHub Actions for ML Pipelines
- Automating ML Workflows with MLflow & Kubeflow

Hands-on Exercise